

DK

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KING'S INVESTMENT FUND · PORTFOLIO CASE STUDY

CashCraft Portfolio Report

A defensive, diversified UK & European equity portfolio that preserved capital and beat the FTSE 100 on every risk-adjusted metric through a post-peak, risk-off market regime.

+3.75%

Outperformance vs FTSE 100

0.85

Portfolio beta

+16.93%

Jensen's Alpha (ann.)

2.30

Diversification ratio

£40,000 notional · 8 holdings · 5 sectors · 63 trading days 2 Mar – 2 Jun 2026 · Benchmark: FTSE 100

Capital preserved while the market fell nearly 4%.

The CashCraft portfolio was built for the King's Investment Fund competition with **£40,000 of notional capital** deployed on 2 March 2026 – three days after the FTSE 100 set an all-time closing high. Over the 63-day window it returned **-0.02%** against the FTSE 100's **-3.77%**, an outperformance of roughly **£1,502**, and did so with a lower drawdown and a positive Sharpe ratio in a market whose Sharpe was deeply negative.

-0.02%Portfolio return vs
-3.77% benchmark**+0.14**Sharpe ratio vs -1.03
benchmark**-5.69%**Max drawdown vs
-8.22% benchmark**1.61**Information ratio (active
efficiency)

What was built

A long-only, buy-and-hold portfolio of eight UK- and European-listed large caps across five ICB sectors and two currency zones (55% GBP, 45% EUR, unhedged). The construction thesis targeted a **defensive beta below 1.0** to compress market-correlated losses and limit the compound wealth erosion caused by volatility drag. No leverage, no derivatives, no rebalancing.

What the results showed

- ▶ The portfolio outperformed the FTSE 100 on **every** standard risk-adjusted measure: Sharpe (+0.14 vs -1.03), Jensen's Alpha (+16.93% annualised), Information Ratio (1.61) and maximum drawdown.
- ▶ The construction thesis held: a realised beta of **0.85** and a diversification ratio of **2.30** pushed portfolio volatility below that of every individual holding.
- ▶ Two high-conviction names – **ASML (+18.89%)** and **BP (+13.48%)** – contributed ~4.9 percentage points, more than offsetting the drag from the other six.

Read with appropriate caution. A 63-day window captures a single market regime and is too short for statistically robust conclusions. The strong risk-adjusted metrics are directionally solid but carry wide confidence intervals, and no ex-ante stress testing was run before deployment. These findings are one encouraging data point, not evidence of persistent alpha.

A high-conviction, quality-growth portfolio with a defensive spine.

The objective was capital growth through selective exposure to structurally advantaged large caps, built to a medium-to-high risk tolerance. The final allocation reflects deliberate active positioning away from the FTSE 100 – notably zero Financials (~19% of the benchmark) in favour of Technology and Industrials where conviction was highest at construction.

Style composition

Style factor	Weight	Representative holdings
Growth	47%	Airbus, ASML, Rolls-Royce, Nordex
Quality	35%	Diageo, SAP, AstraZeneca
Value	18%	BP
Thematic overlay	–	Energy security & defence

Constraints observed throughout

- ▶ **Geography:** UK & Western Europe only (4 exchanges).
- ▶ **Asset class:** direct equities only – no ETFs, funds or derivatives.
- ▶ **Leverage:** none; fully funded by the £40,000 base.
- ▶ **Currency:** GBP base with 45% unhedged EUR exposure.
- ▶ **Concentration:** no single name above 20% of NAV at inception.
- ▶ **Structure:** single lump-sum entry, buy-and-hold, no rebalancing.

Why low volatility was the design goal. Geometric compound returns are reduced below the arithmetic mean by roughly $\frac{1}{2}\sigma^2$ – the well-documented **variance drag**. Minimising volatility is therefore a precondition for compounding wealth, which is why the portfolio was engineered around a sub-unity beta rather than maximum expected return.

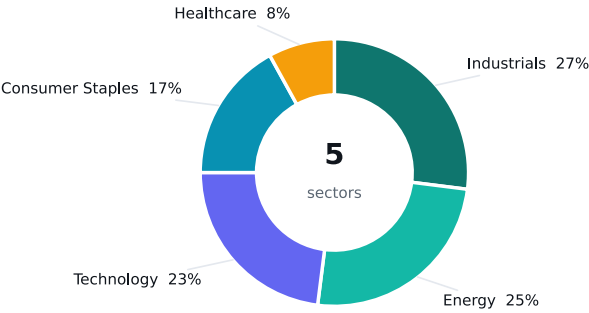
Eight holdings, five sectors, two currency zones.

£40,000 was deployed in full on 2 March 2026 using fractional shares to match target weights exactly. Euro-denominated holdings were converted to GBP at the 2 March rate (1 EUR = 0.8739 GBP, Yahoo Finance).

Ticker	Company	Sector	Entry (GBP)	Cost	Weight
BP.L	BP	Energy	£4.78	£7,200	18%
DGE.L	Diageo	Consumer Staples	£16.63	£6,800	17%
AIR.PA	Airbus	Industrials	£161.01	£6,000	15%
ASML.AS	ASML	Technology	£1,078.03	£5,200	13%
RR.L	Rolls-Royce	Industrials	£13.34	£4,800	12%
SAP.DE	SAP	Technology	£149.40	£4,000	10%
AZN.L	AstraZeneca	Healthcare	£155.42	£3,200	8%
NDX1.F	Nordex	Energy (wind)	£37.68	£2,800	7%
Total invested				£40,000	100%

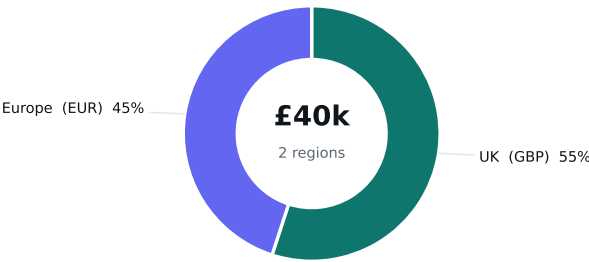
Sector allocation

Five ICB sectors, none above 27%



Geographic split

55% UK (GBP) · 45% EU (EUR, unhedged)



0.118

Herfindahl-Hirschman Index
(moderately concentrated, below 0.25)

50%

Top-3 holdings (BP, Diageo, Airbus)

7.33

Effective number of positions (near equal-weight)

Each position earned its place.

Holdings were chosen on fundamental quality, valuation relative to sector peers, identifiable near-term catalysts and durable competitive advantage. Returns shown are GBP-adjusted over the 63-day window.

ASML.AS ASML Holding · **+18.89%**

Technology

Sole manufacturer of EUV lithography – a literal monopoly on the equipment needed for leading-edge chips. Record Q4 order intake (€13.2bn) and the AI-driven semiconductor capex cycle drove the portfolio's strongest contribution.

Monopoly moat

Fwd P/E ~43x

ROE 52%

Largest contributor

BPL BP · Energy **+13.48%**

Cheap integrated energy major (fwd P/E ~8–9x, ~5% yield) held as the value anchor. The US-Iran energy shock lifted upstream revenues, and BP's negative correlation with the industrials and tech names provided an emergent internal hedge.

Value + hedge

EV/EBITDA ~5x

β -0.22

Natural diversifier

RR.L Rolls-Royce · Industrials **-5.42%**

Turnaround story – FY2025 operating profit +40%, 17.3% margin, aftermarket recovery and a second growth engine from data-centre power demand. Duopoly position in wide-body engines with a recurring LTSA backlog.

Turnaround

Op margin 17.3%

FCF £3.3bn

AIR.PA Airbus · Industrials **-5.71%**

Structural growth with an 8,500+ aircraft backlog – nearly a decade of visibility. A320neo rate ramp and European rearmament provide the catalysts; effective duopoly with Boeing.

Structural growth

Backlog ~10 yrs

Trailing P/E ~26x

SAP.DE SAP · Technology **-4.06%**

Cloud transition inflecting to profitability – operating margin expanded from 13.7% to 25.2% in a year. ERP lock-in and the mandatory S/4HANA migration provide structural demand.

Cloud transition

Gross margin 74%

Cloud +27% YoY

NDX1.F Nordex · Energy **-7.75%**

(renewables)

Cyclical recovery in onshore wind – FY2025 earnings turnaround, EBITDA margin doubling, €17.0bn order book. Adds a clean-energy / ESG dimension alongside BP.

Cyclical recovery

Order book €17bn

57 GW installed

DGE.L Diageo · Consumer Staples **-8.08%**

Mean-reversion play on a de-rated premium-spirits leader (P/E ~15x vs 5-yr avg ~25x) with a £625m cost programme. The anticipated re-rating did not materialise inside the window.

Mean reversion

Yield 3.9%

Gross margin 60%

AZN.L AstraZeneca · Healthcare **-13.36%**

Defensive anchor (β 0.19) with a deep oncology pipeline. Underperformed on a broad pharma de-rating rather than any company-specific deterioration – the portfolio's weakest selection.

Defensive anchor

Rev +8.6%

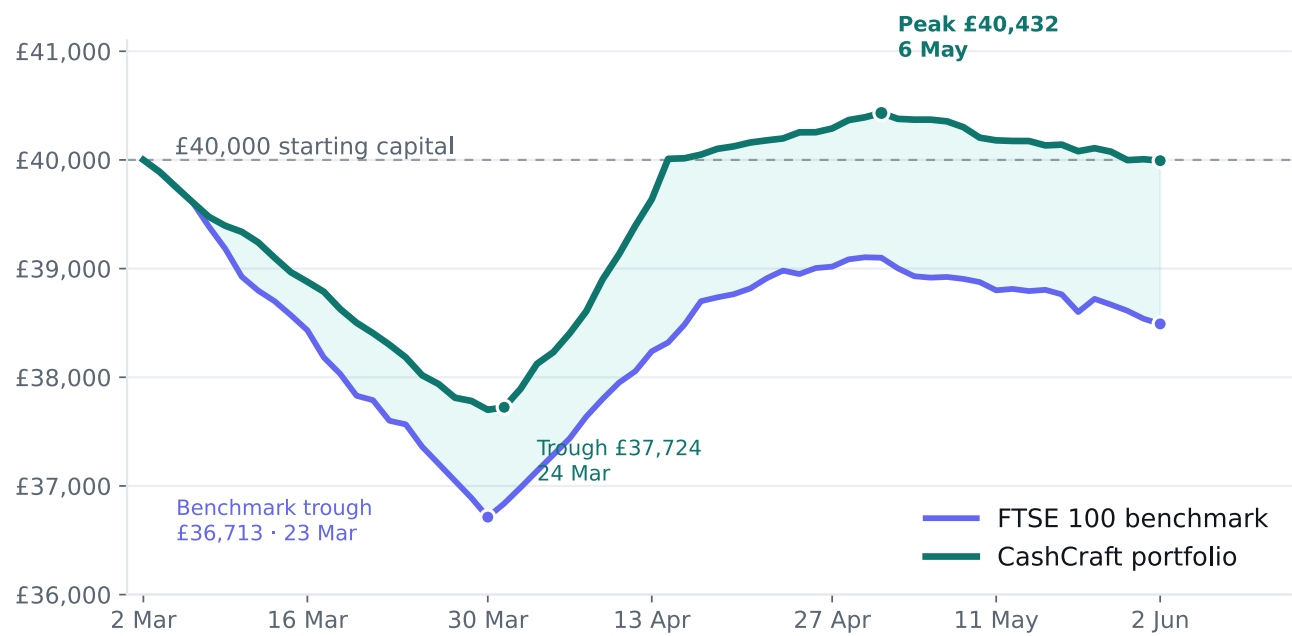
Low beta

Shallower falls, faster recovery.

Both £40,000 positions entered immediately after the FTSE 100's all-time high and fell together into a late-March trough. The portfolio's decline was shallower, it reclaimed its starting capital by mid-April and set a new high of £40,432 on 6 May. The benchmark position never recovered its opening value.

Cumulative value - CashCraft vs FTSE 100

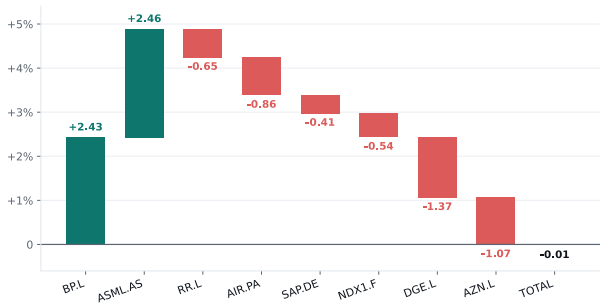
£40,000 deployed 2 Mar 2026 · 63 trading days · GBP



The portfolio troughed one day after the benchmark and 1.0pp shallower, then diverged above its starting capital for most of April–May. **Closing gap: ~£1,502 in the portfolio's favour.**

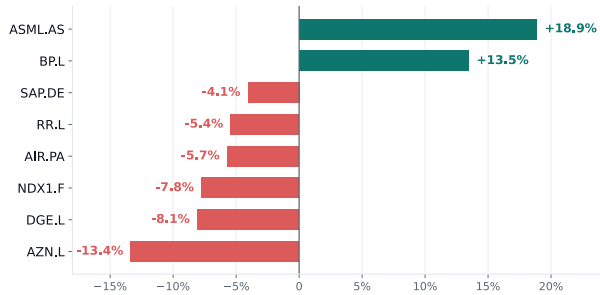
Return contribution by holding

Weight × return, waterfall to portfolio total



Individual stock returns

GBP-adjusted, 63 trading days



Annualised return, excess return and alpha

Scaling the daily series to a 252-day year, the portfolio's positive drift stands in clear contrast to the benchmark's negative drift. Alpha is positive in both period and annualised views – the return the portfolio earned *beyond* what its 0.85 beta alone would explain.

Series	Mean daily	Annualised
CashCraft portfolio	+0.0242%	+6.28%
FTSE 100 benchmark	-0.0557%	-13.11%
Difference	+0.0799%	+19.39%

+16.93%

Jensen's Alpha (annualised) – risk-adjusted outperformance beyond CAPM

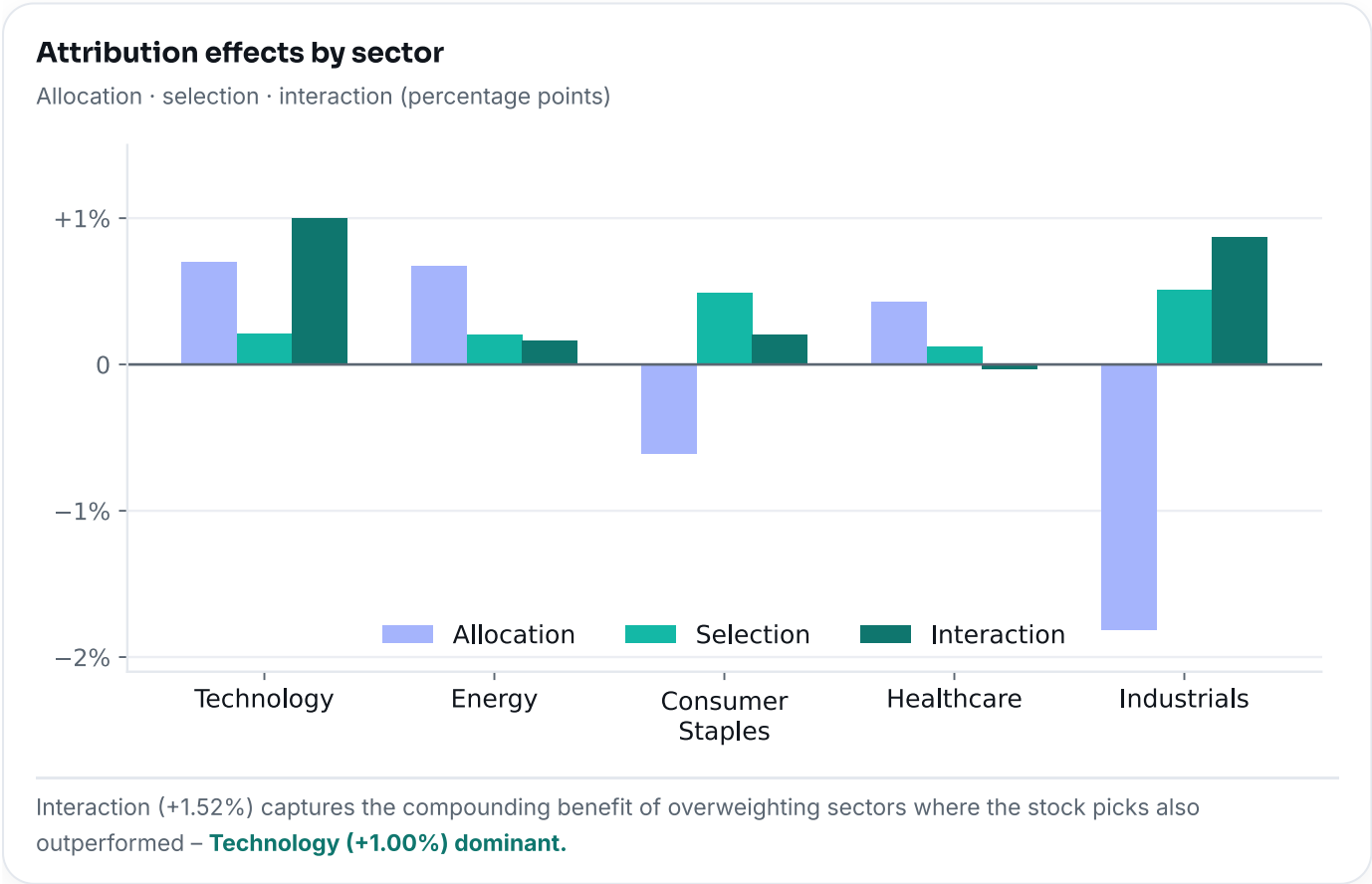
+3.06%

Jensen's Alpha (period) – attributable to selection, not market exposure

Reconciling two views. A positive arithmetic daily mean (+0.024%) compounded to an essentially flat cumulative return (−0.02%). The gap is **variance drag** – at 1.10% daily volatility it accumulated meaningfully over 63 days and absorbed the positive drift. This is precisely the effect the low-volatility construction was designed to minimise.

Performance attribution (Brinson-Hood-Beebower)

Decomposing the +3.75% active return, outperformance was driven by **stock selection (+2.18%)**, not sector allocation (+0.05%). The portfolio's picks beat the FTSE 100 sector benchmark in every sector.

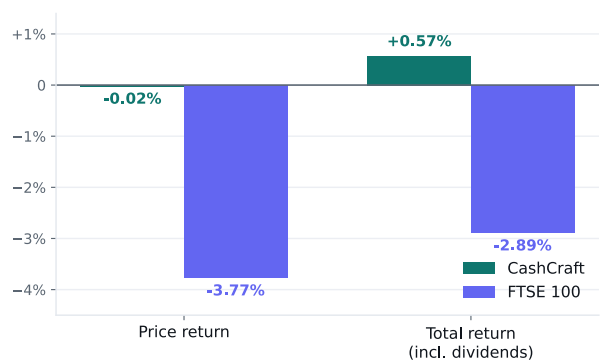


Total return including dividends

On a price basis the portfolio was flat; adding estimated dividend income (pro-rated over the window) shifts it to a modest **+0.57%** – capital preserved and a small positive real return.

Price vs total return

Including pro-rated dividend income



Metric	Portfolio	FTSE 100
Price return	-0.02%	-3.77%
Est. dividend return	+0.59%	+0.88%
Est. total return	+0.57%	-2.89%

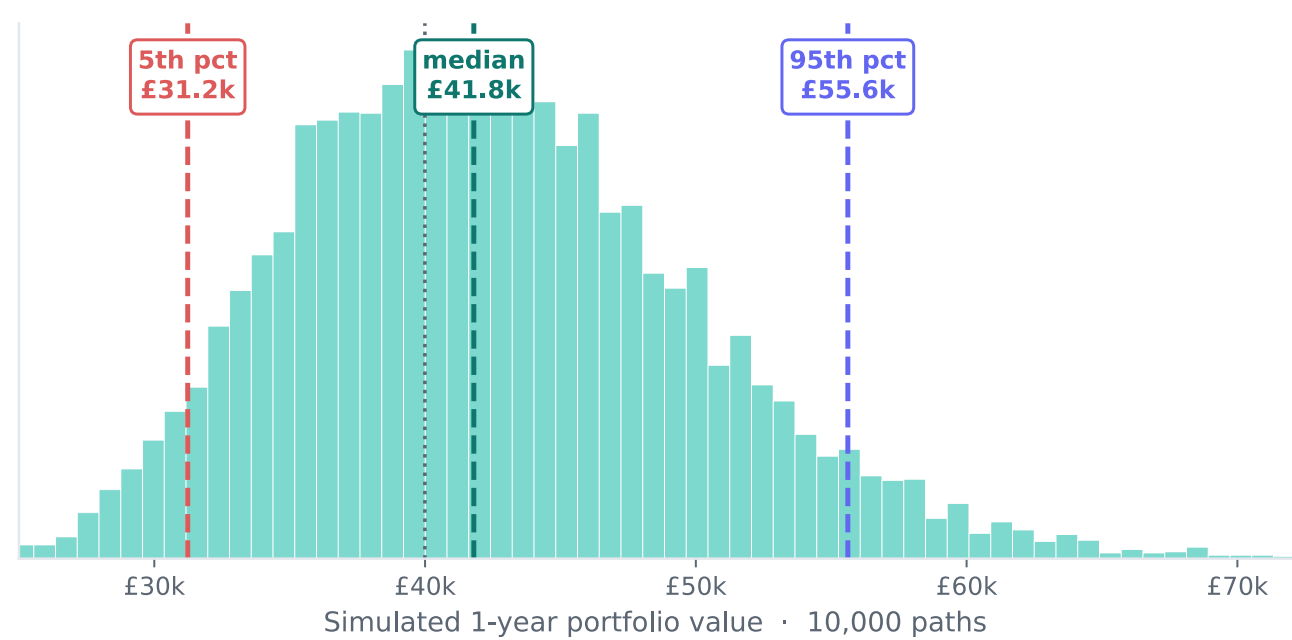
The benchmark's higher yield (3.5% vs 2.36%) narrows the active gap slightly on a total-return basis – a direct consequence of the portfolio's growth tilt toward lower-yielding names such as ASML and SAP.

Forward view – Monte Carlo simulation

10,000 one-year paths were simulated under geometric Brownian motion using the window's observed daily mean and volatility. The distribution is skewed to the upside with a **60.9% probability of profit**.

Simulated 1-year terminal value

10,000 GBM paths · 252 trading days



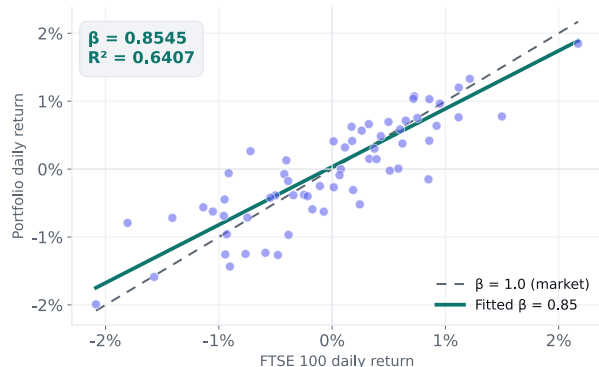
Mean £42,439 (+6.1%) · median £41,801 (+4.5%) · 5th pct £31,234 (-21.9%) · 95th pct £55,614 (+39.1%). **Illustrative only** – assumes the 63-day return distribution persists and normal tails, which understate extreme outcomes.

Lower systematic risk, contained tails.

Total volatility was marginally higher than the benchmark (17.55% vs 16.44%), but that excess is **idiosyncratic** – the price of an 8-stock portfolio. The **systematic** component (14.05%) that drives correlated drawdowns was materially lower, a direct result of the 0.85 beta.

Beta regression · daily returns

Slope of the fitted line = portfolio beta



$\beta = 0.85$ captured ~85% of market moves; $R^2 = 0.64$ means ~36% of variance was stock-specific – consistent with an active, tilted portfolio.

14.05%

Systematic volatility (vs benchmark 16.44%)

10.43%

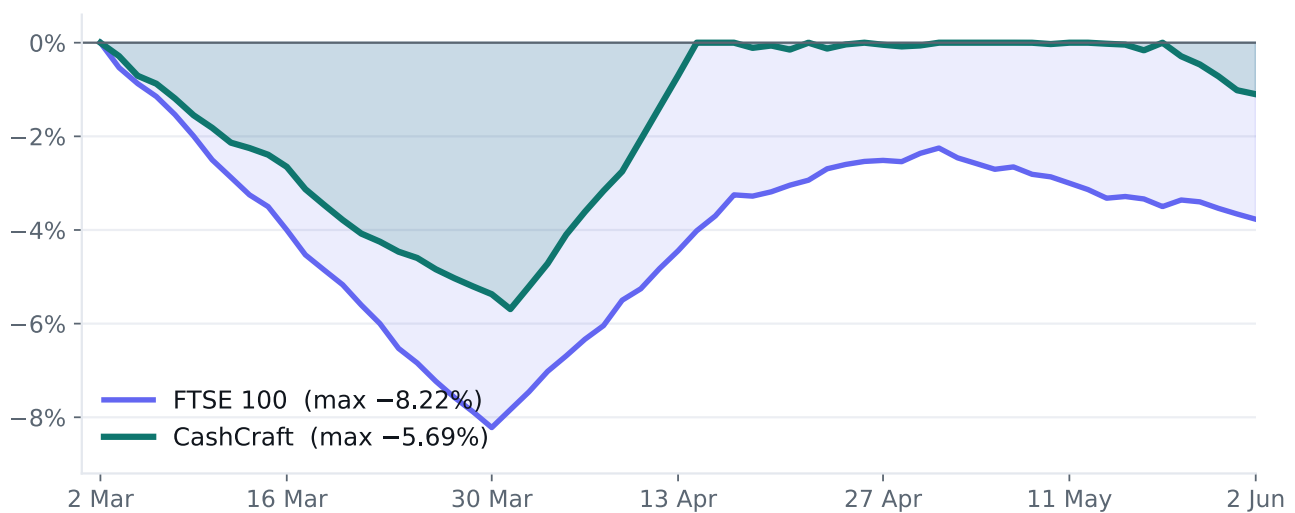
Idiosyncratic volatility (diversifiable)

~80 / 20

Variance split: systematic / stock-specific

Drawdown · CashCraft vs FTSE 100

Peak-to-trough decline through the window



The shaded advantage is widest at the March trough and the late-May pullback. Maximum drawdown was **2.53pp shallower** than the benchmark's.

Value at Risk & tail behaviour

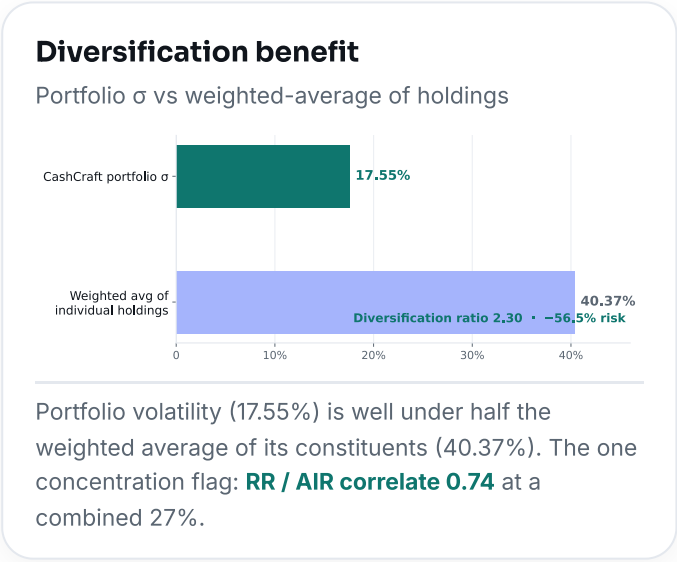
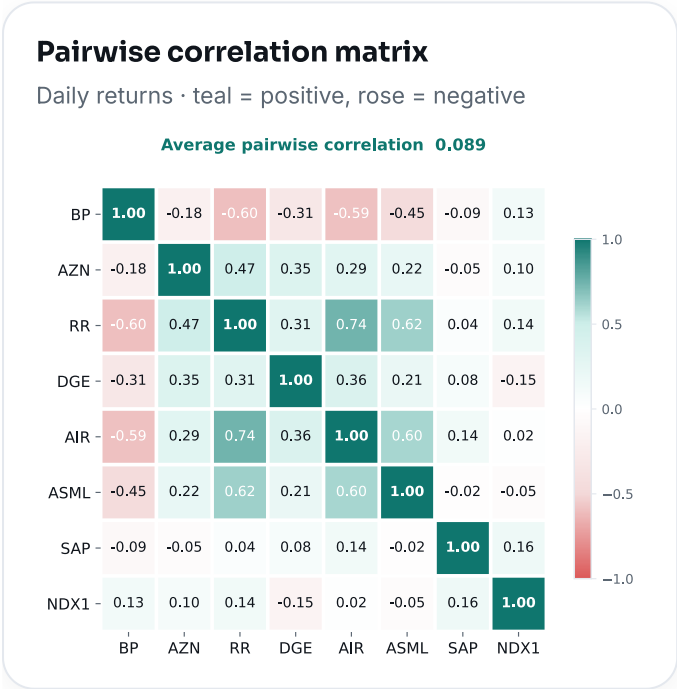
Under the parametric method the portfolio looks marginally riskier (its total σ is higher). Under the **historical** method the result reverses – the portfolio's empirical tails were thinner, i.e. its worst days were less extreme than its volatility implied.

Confidence	Method	Portfolio	FTSE 100	Portfolio £ at risk
95%	Parametric	1.78%	1.75%	£712
95%	Historical	1.41%	1.54%	£565
99%	Parametric	2.53%	2.44%	£1,011
99%	Historical	2.45%	2.50%	£980

Tails were contained. The portfolio's worst single day (–2.52%) was shallower than the benchmark's (–2.75%), and this advantage held across all five worst days. CVaR (expected shortfall) at 95% was 2.15% vs 2.27% – when the threshold was breached, the average loss was smaller too.

Diversification

An average pairwise correlation of just **0.089** – far below the 0.30–0.50 typical of a single-region equity book – drove a diversification ratio of 2.30, cutting weighted-average risk by 56.5%. BP's negative correlation with the non-energy names was the key emergent hedge.



Outperformance on every metric that matters.

Measured against the FTSE 100 across return, risk and risk-adjusted dimensions, the portfolio led on all but one line – total volatility, where the gap is idiosyncratic rather than systematic.

Metric	CashCraft	FTSE 100	Differential
Cumulative return (63d)	-0.02%	-3.77%	+3.75pp
Annualised return	+6.29%	-13.11%	+19.40pp
Annualised volatility	17.55%	16.44%	+1.11pp
Beta	0.85	1.00	-0.15
Maximum drawdown	-5.69%	-8.22%	+2.53pp
Sharpe ratio	+0.14	-1.03	+1.17
Treynor ratio	+0.0287	-0.1694	-
Jensen's Alpha (ann.)	+16.93%	-	-
Information ratio	+1.61	-	-
Tracking error	12.43%	-	-
M ² (risk-matched)	+19.24%	-	-

Active management effectiveness

- ▶ **Hit ratio 57.1%** – beat the benchmark on 36 of 63 days.
- ▶ **Symmetric win/loss (±0.55%)** – gains from frequency, not oversized bets.
- ▶ **IR 1.61** – active risk was efficiently rewarded (above 1.0 is exceptional).

The one weakness. Total volatility was 1.11pp higher than the benchmark – the cost of holding 8 names rather than 100. It reflects diversifiable idiosyncratic risk, not greater market sensitivity.

A two-speed, risk-off regime.

The window was defined by escalating US-Iran tensions after the FTSE 100's 27 February peak. Energy and defence outperformed while the broad index fell – a divergence the portfolio's sector mix captured on both sides.

3.75%
BoE base rate held all window (6 prior cuts)

3.3%
CPI inflation in March, energy-led

+0.1%
Q4 2025 UK GDP – near-stagflation

The oil-price spike directly lifted BP (+13.48%) while triggering a risk-off rotation out of growth, consumer and industrial equities – explaining much of the FTSE 100's decline and the weakness across the portfolio's non-energy holdings.

Tensions acknowledged, not hidden.

The portfolio was not built on an ESG screen – it holds an oil major, two defence-exposed industrials, a drinks giant and a wind-turbine maker. A credible report names the tensions rather than sweeping them aside.

Holding	MSCI	Ethical tension	Counter-argument
Diageo	AAA	Alcohol-related harm	Leader in responsible-drinking programmes
ASML	AAA	Conflict minerals in supply chain	Enables energy-efficient chips
SAP	AA	Data privacy	Software for corporate carbon accounting
Nordex	AA	Land use, bird strikes	Direct renewable-energy contribution
Airbus	BBB	Dual-use technology	Adheres to EU arms controls
AstraZeneca	A	Drug pricing / access	Breakthrough oncology, patient-access schemes
BP	A	Fossil fuels, climate lobbying	Net-zero commitment, hydrogen investment
Rolls-Royce	BB	Defence (military engines)	Developing small modular nuclear reactors

Over a longer horizon, BP's weight could be reduced and reallocated toward cleaner-energy and ESG-aligned objectives.

What worked, what didn't, what changes next time.

What worked well

- ▶ The low-volatility thesis was empirically validated – beta 0.85 compressed the market's -3.77% to an effective -3.22%, with selection recovering the rest.
- ▶ Diversification was the standout structural feature – a 2.30 ratio and 0.089 average correlation from only 8 names.
- ▶ The quantitative framework (daily FX-adjusted tracking, Sharpe/alpha/IR) enabled genuine risk-adjusted attribution rather than raw P&L observation.
- ▶ The 45% European sleeve added sectors absent from the FTSE 100 – ASML's +18.89% would not have existed under a UK-only mandate.

Mistakes made

- ▶ The 23% Technology weight (vs ~2% benchmark) was the largest uncompensated active bet and sat awkwardly against a low-volatility thesis.
- ▶ No ex-ante stress testing or backtesting before deployment – a time constraint that left tail behaviour unknown at capital commitment.
- ▶ No cross-asset sleeve despite a directional macro view – a 5–15% gold allocation would likely have trimmed drawdown by 1–2pp.
- ▶ The RR / AIR pair ($\rho = 0.74$, combined 27%) was a hidden single bet on European aerospace, flagged only ex-post.

Behavioural honesty. Post-hoc reflections such as "I should have bought gold" are hindsight bias – the FTSE 100 was at all-time highs three days before deployment. The defensible framing is "I assigned a higher probability to a risk-off regime than the market was pricing," not certainty. Separating process from outcome matters: a good result does not validate a biased process.

Future improvements

- ▶ Make ex-ante stress testing (2008, March 2020, 2022) and Monte Carlo a mandatory pre-deployment gate with explicit acceptance thresholds.
- ▶ Add a cross-asset sleeve (gold, gilts or a vol-target) to break the equity correlation floor.
- ▶ Screen pairwise correlations at construction, not only in review.

Objectives met; a defensible single data point.

Objective	Outcome	Evidence
Diversified UK/EU equity portfolio	Achieved	8 holdings, 5 sectors, 2 regions · diversification ratio 2.30 · avg correlation 0.089
Defensive beta below 1.0	Achieved	Realised β 0.85 · market-attributed loss -3.22% vs benchmark -3.77%
Minimise volatility drag	Partial	Systematic σ 14.05% below benchmark; total σ marginally higher on idiosyncratic risk

Beyond its stated targets the portfolio outperformed the FTSE 100 on every risk-adjusted metric – a positive Sharpe against a deeply negative market Sharpe, Jensen's Alpha of 16.93% annualised, an Information Ratio of 1.61 and a shallower maximum drawdown. It recovered its starting capital within six weeks; the benchmark did not recover within the window.

Future outlook

The current composition – overweight energy and technology, zero financials, beta 0.85 – implies a continued bias toward capital preservation over benchmark tracking. It is better positioned for continued volatility than for a sharp, financials-led recovery, where the defensive tilt would lag. That is consistent with the thesis: the strategy was designed to compound steadily across regimes, not to maximise returns in a rising market. A longer holding period spanning both drawdown and recovery would provide the more complete test.

Bottom line. The portfolio preserved capital and beat its benchmark on every risk-adjusted measure through a genuine risk-off regime, validating a deliberate low-volatility, diversified construction. The honest caveat – a 63-day, single-regime window – is stated throughout: this is one encouraging, defensible data point rather than proof of persistent alpha.

Data sources & methodology

Daily closing prices, FX and index data: Yahoo Finance (2 Mar – 2 Jun 2026). Fundamentals: company FY2025/26 reports, StockAnalysis.com, Investing.com (TTM basis). Risk-free rate: UK 3-month Treasury Bill, 3.84% annualised (Bank of England). Attribution uses the Brinson-Hood-Beebower framework. All figures traceable to the CashCraft Portfolio Master workbook. Charts rebuilt for this report; the cumulative-value and drawdown series are reconstructed from the documented daily valuations and inflection points. Selected references: Sharpe (1966, 1994), Jensen (1968), Treynor (1965), Brinson-Hood-Beebower (1986), Markowitz (1952), Grinold & Kahn (1999).